Course Description

BSCS

Course Title	Artificial Intelligence
Instructor	Husnain Haider
Summary of the course	This course will introduce the basics of artificial intelligence (AI), its scope and application domain
Objectives	AI is a field that studies how to realize the intelligent human behaviors on a computer. The ultimate goal of AI is to make a computer that can learn, plan, and solve problems autonomously. Understand the various searching techniques, example problems-game playing techniques. Apply these techniques in applications which involve perception, reasoning and learning. Explain the role of agents and how it is related to environment and the way of evaluating it and how agents can act by establishing goals. Acquire the knowledge of real world Knowledge representation. Analyze and design a real world problem for implementation and understand the dynamic behavior of a system.
Structure	Lectures - Lab
Credit Hours	3(2+1)
Prerequisites	None
Textbook	Artificial Intelligence: A modern Approach by Russell and Norvig
Reference Material	Artificial Intelligence Illuminated by Ben Coppin

Lecture	Topics
1	**Introduction to Artificial Intelligence**: History, Goals, Applications, Different AI Approaches, Turing Test, Cognitive models, Syllogism, Rational Agents **Intelligent Agents**, The Concept of Rationality, The Nature of Environments, The Structure of Agents
2	**Solving Problems by Searching**, Problem-Solving Agents, Problem Formulation, Search spaces, State space, Search Trees, Initial and Goal State, Successor Function, Objective Function, Goal Test, Goal Path, Example Problems (Missionaries & Cannibals, Wolf Duck and Farmer, 8-Puzzle, Tic-Tac-Toa, 8-Queen, TSP)
3	**Uninformed/Blind Search**: Breadth First Search (BFS), Depth First Search (DFS), Time Complexity, Space Complexity, Completeness, Optimality, Depth Limited Search, Uniform Cost Search, IDS, Algorithm Complexity
4	**Informed searches**: Heuristics, Heuristic Functions and Properties, Best First Search, Greedy Search
5	**Informed Searches**: A* Search, A* Search examples

6	**Local Search** : Hill Climbing, Local minima and maxima, Variants of Hill Climbing Algorithm
7	**Adversarial Search**: Game Trees, Minimax, Alpha beta pruning
8	**Genetic Algorithms**: Representation, Fitness, Selection, Mutation and Cross over, Solution of 8-queens , TSP through Genetic Algorithms
	Mid Term
9	**Introduction to Machine Learning (ML)**: ML classification, Classical vs ML approach, Naïve Bayes for classification example
10	**Curve Fitting**: Polynomial Curve fitting, Problem Representation, Error Function and root mean square error
11	**Curve Fitting**: Polynomial Curve fitting, Generalization and Over-fitting, How to avoid over-fitting
12	**Principle Component Analysis (PCA)**,: Background, Data representation and optimal principle component derivation
13	**Principle Component Analysis (PCA)**,: Classification with PCA
14	**Clustering with K-means:** Examples
15	**Introduction to Artificial Neural Network (ANN)**: Forward Propagation, Back Propagation
16	**Revision**